AQA

SPECIMEN MATERIAL

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Centre number | | | | | | Candidate number | | | | |

Surname _____

Forename(s) _____

Candidate signature _____

Functional Skills Level 2 MATHEMATICS (8362)

Paper 1 Non-Calculator Paper

Specimen paper

Time allowed: 30 minutes

Materials

For this paper you must have:

- mathematical instruments.

You must **not** use a calculator.



Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 20.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.

Advice

- In all calculations, show clearly how you work out your answer.

Do not write outside the box

Section A

Answer **all** questions in the spaces provided.

1 What is $\frac{3}{5}$ as a decimal?

Circle your answer.

[1 mark]

0.06 0.35 0.53 0.6

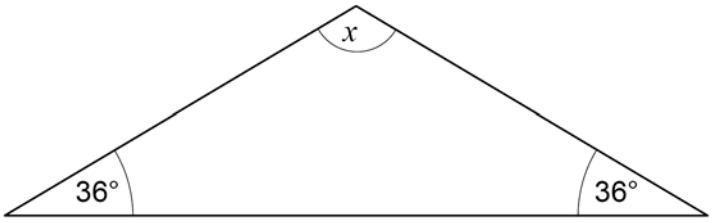
2 Work out 7 – 2.835

[1 mark]

Answer _____

3 Work out the size of angle x in this triangle.

[2 marks]



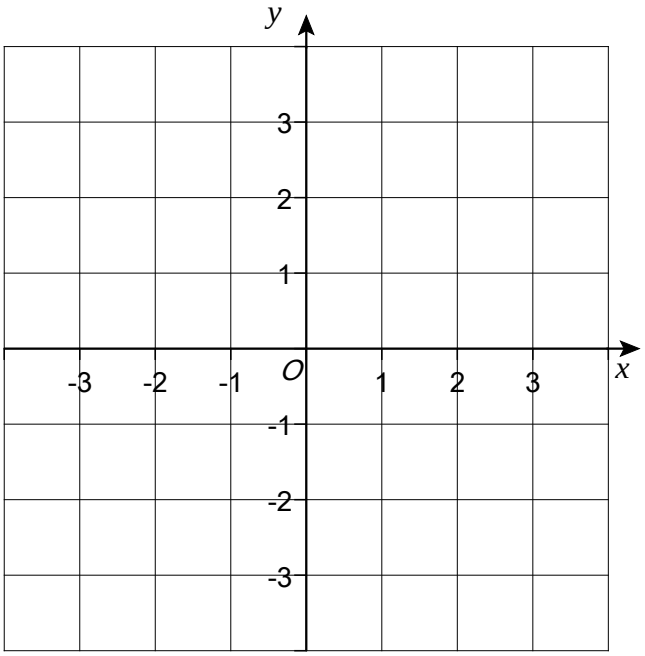
Not drawn accurately

Answer _____ °

Do not write outside the box

4 Plot the point (–3, 2) on this grid

[1 mark]



5 Work out $25 – 2 \times 3^2$

[2 marks]

Answer _____

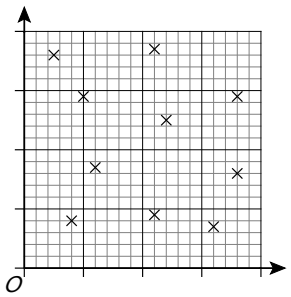
Turn over for the next question

7

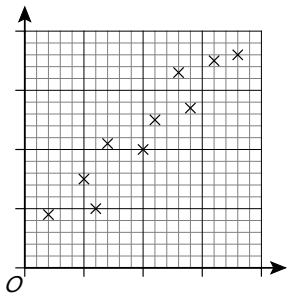
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6 **A**, **B**, **C** and **D** are scatter diagrams.

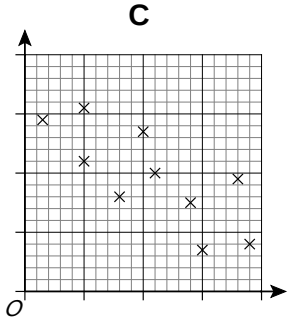
A



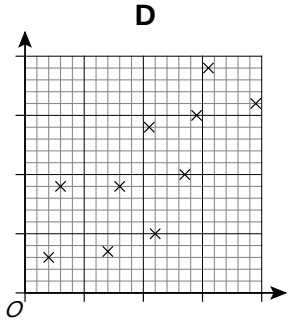
B



C



D



Which diagram shows negative correlation?
Circle your answer below.

[1 mark]

A **B** **C** **D**

1

Do not write outside the box

Turn over for the next question

DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

7 **Market stall**

Lisa makes fudge and sells it on a market stall.

7 (a) Lisa makes fudge using these ingredients.

sugar	500	grams
butter	85	grams
evaporated milk	175	grams
milk	150	millilitres
flavouring	20	millilitres

This makes enough fudge for 4 boxes.
Lisa wants to make enough fudge for 48 boxes.
She already has 1.4 kilograms of sugar.
Sugar is sold in 1 kilogram bags.

How many bags of sugar does she need to buy?

[5 marks]

Answer _____

Do not write outside the box

7 (b) Lisa sells the fudge in three flavours: strawberry, vanilla and mint.

She looks at the number of boxes of each flavour she has sold in the last few weeks.

Strawberry	60
Vanilla	140
Mint	40

This week, Lisa makes 48 boxes.

Using the data above, how many boxes of each flavour should she make?
You **must** show your working.

[4 marks]

Strawberry _____

Vanilla _____

Mint _____

Question 7 continues on the next page

9

Do not write outside the box

7 (c) Lisa has to drive 50 miles to the market.

The market starts at 9 am

She needs to arrive at least half an hour before the market starts.

She leaves home at 7.10 am

Lisa says,

"If I drive at an average of 40 miles per hour I will be there in time."

Is she correct?
You **must** show your working.

[3 marks]

3

END OF QUESTIONS

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Functional Skills Level 2 MATHEMATICS (8362)

Paper 2 Calculator Paper

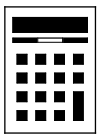
Specimen paper

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a calculator
- mathematical instruments.



Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a π button, take the value of π to be 3.142

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Pages	Mark
2-3	
4–5	
6–7	
8–9	
10–11	
12–13	
14–15	
16–17	
18–19	
20	
TOTAL	

Do not write outside the box

Section A

Answer **all** questions in the spaces provided.

1 A set of numbers is 4 6 6 7 8 8 8 9

Circle the mode.

[1 mark]

5 7 7.5 8

2 Work out $\frac{9.386 + 20.904}{2.5}$

Give your answer as a decimal

[1 mark]

Answer _____

3 The probability of event A happening is 0.15

Work out the probability of event A **not** happening.

[1 mark]

Answer _____

Do not write outside the box

4 Circle the calculation that increases £260 by 17%

[1 mark]

260 + 0.17 260 × 0.17 260 + 1.17 260 × 1.17

5 Write in digits four hundred and three thousand, seven hundred and twenty.

[1 mark]

Answer _____

6 Work out $5\frac{3}{4} - 1\frac{1}{8}$

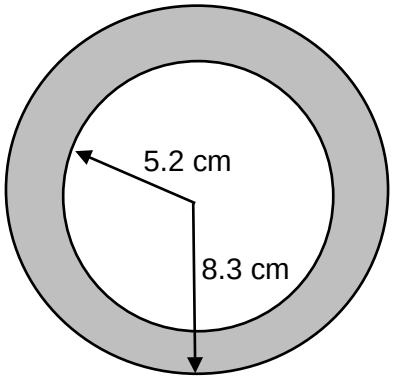
[1 mark]

Answer _____

6

Do not write outside the box

7 A circle of radius 5.2 cm is inside a circle with radius 8.3 cm



Not drawn accurately

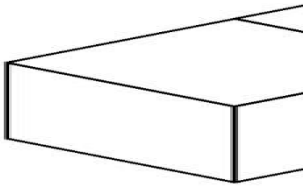
Work out the shaded area.

[2 marks]

Answer _____ cm^2

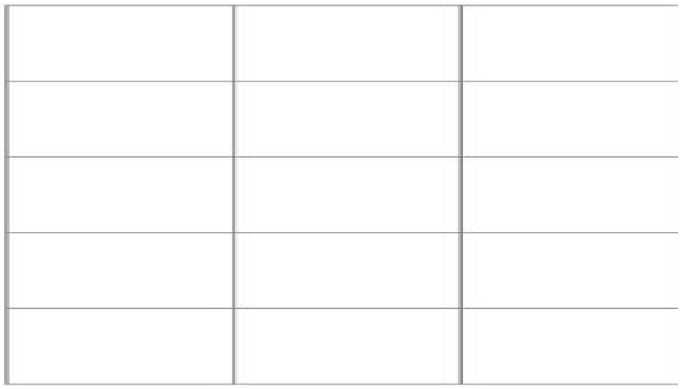
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8 Each of the **four** cubes in this L-shape has side length 1 centimetre.



On this centimetre grid draw a plan view of the L-shape.

[1 mark]



9 Work out the percentage decrease from 5200 to 4108

[3 marks]

Answer _____ %

6

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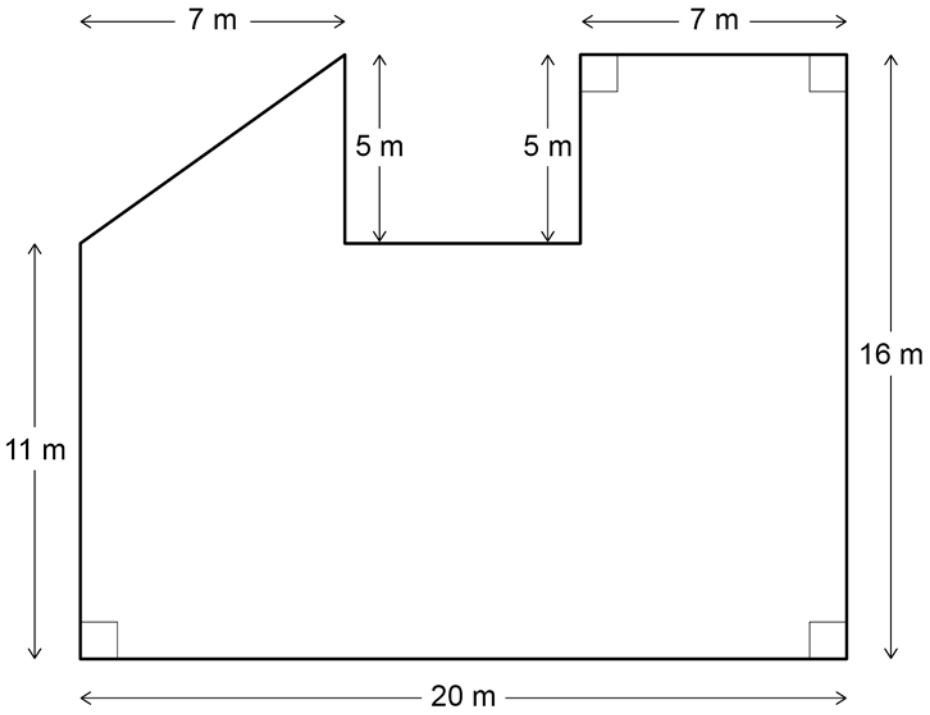
Section B

Answer **all** questions in the spaces provided.

10 **Playground**

Levi works for a company that designs and builds playgrounds.

Here is a sketch of the plan for a new playground.



10 (a) Work out the area of the playground.

[3 marks]

Answer _____ m^2

Do not write outside the box

10 (b) The surface of the playground will be covered with rubber chips and resin.

Levi uses 14 kg of rubber chips per square metre

He also uses resin, in the ratio

mass of rubber chips : mass of resin = 5 : 1

The resin is supplied in 25 kg tubs.

How many tubs does Levi need?

[5 marks]

Answer _____

8

Do not write outside the box

10 (c) The table shows the items needed for the playground.

	Space needed
1 climbing frame	6 m by 6 m square
1 swing set	10 m by 4 m rectangle
1 sandpit	semicircle with radius 6 m
2 rockers	each 2 m by 2 m square

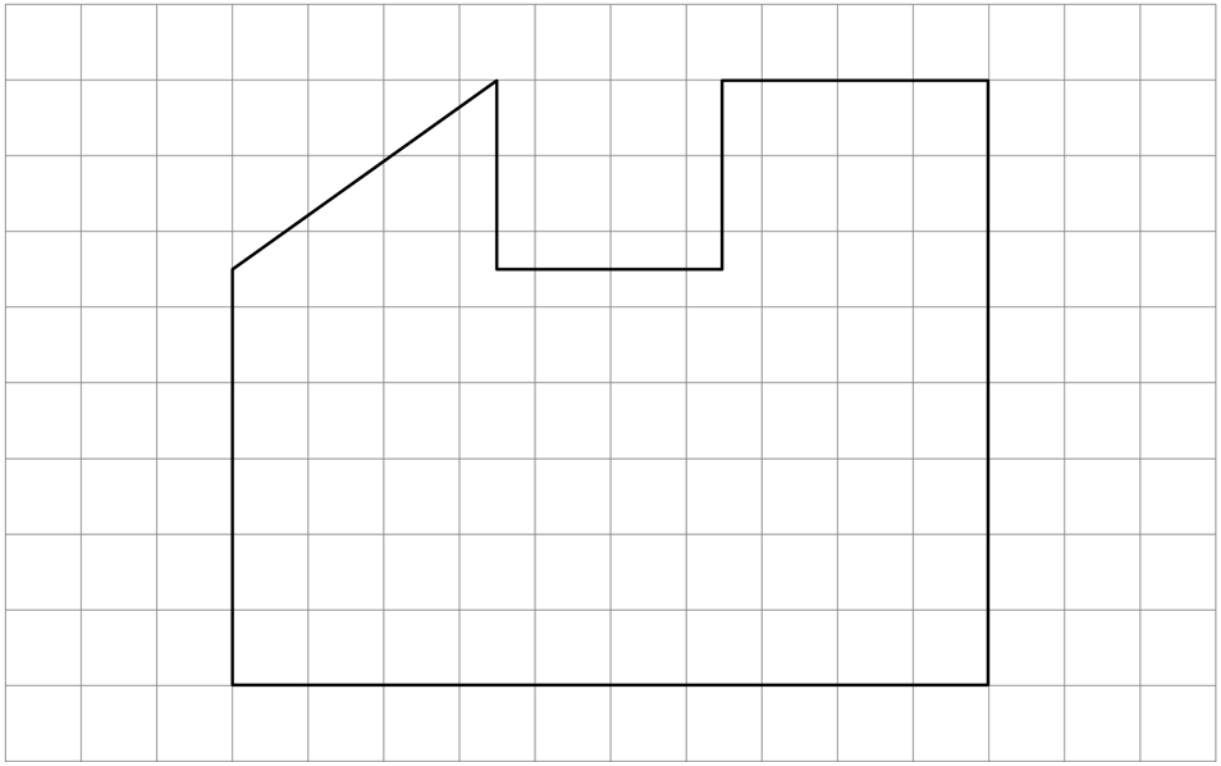
The playground is drawn to a scale of 1 to 200 on a centimetre grid.

On the grid, design a possible playground.

[5 marks]

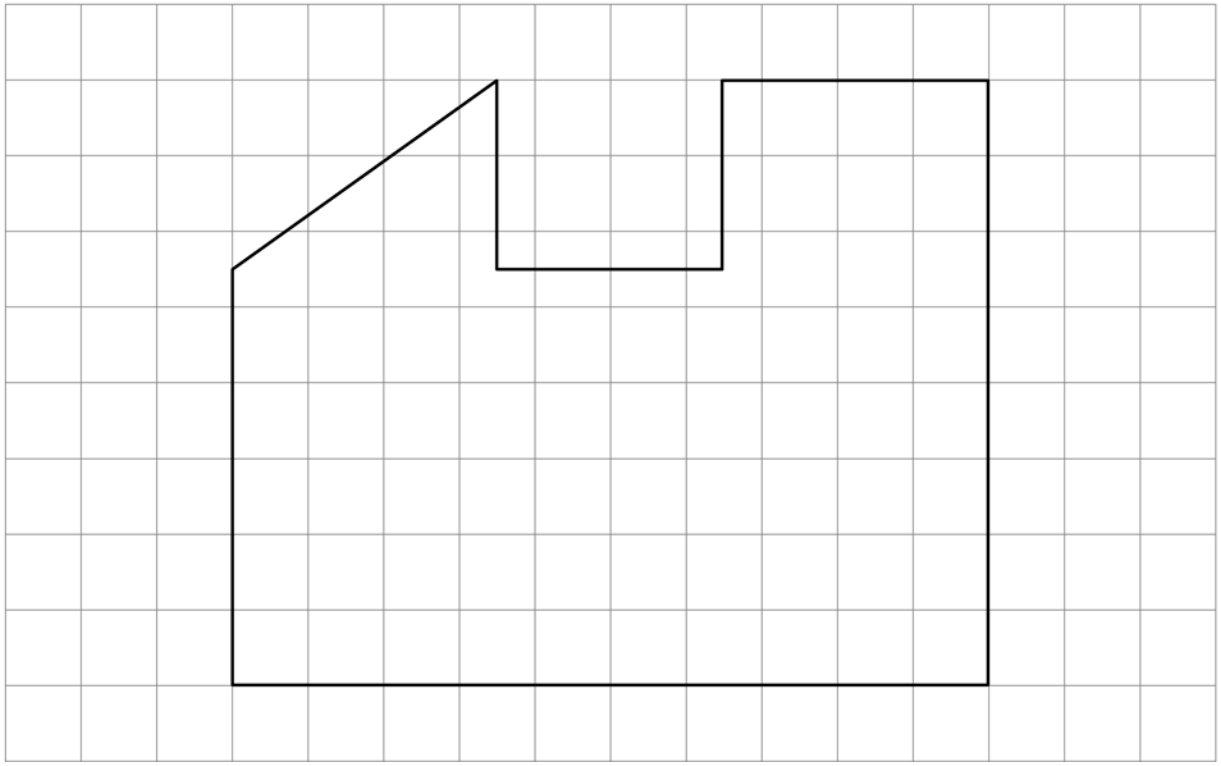
Practise on this grid.

Scale: 1 to 200



Put your answer on this grid.

Scale: 1 to 200



Do not write outside the box

5

Do not write outside the box

11 Fundraising

Padma is organising a fundraising event for a charity.

The event will include

a 3-course meal

a singer.

11 (a) Padma is working out how much to charge for a ticket to the event.

The table shows the costs she expects to have.

Hire of venue	£1660
Meal	£14.25 per person
Fee for singer	£400
Other costs	£350

Padma expects 230 people to attend the event.

She wants to make a profit of at least £5000 to give to the charity.

Work out the smallest amount she should charge for a ticket.
Give your answer to a suitable degree of accuracy.
You **must** show your working.

[6 marks]

Do not write outside the box

Answer £ _____

Question 11 continues on the next page

6

Do not write outside the box

11 (b) Padma wants to have a number of flyers printed to promote the event.

She sees these adverts for two printing companies

North Printers
Flyers: 200 for £3.25

Speedy Printers
Flyers: 1000 for £18
12.5% discount on orders over £70

It would cost £97.50 to have the flyers printed at North Printers.

How much would it cost to have the same number of flyers printed at Speedy Printers?

[5 marks]

Answer £ _____

Do not write outside the box

11 (c) At the event, Padma sells 800 raffle tickets for £2 each.

Altogether, the people on one table spend £110 on raffle tickets.

Padma tells the people on the table,

"The probability that someone on this table wins the first prize is more than 5%"

Is she correct?

You **must** show your working.

[3 marks]

Turn over for the next question

8

Do not write outside the box

12 **Quiz**

Eve and Stefan each take part in a quiz every week.

They look at their scores in the first 12 weeks that the quiz takes place.

Here is a summary of the data for Eve.

Range	15
Mean	41.25

The frequency table shows the data for Stefan.

Score	Frequency
37	1
38	1
39	0
40	4
41	2
42	0
43	4

Do not write outside the box

12 (a) Stefan says,

"My scores were more consistent."

Is he correct?
Give a reason for your answer.
You **must** show your working.

[2 marks]

12 (b) Eve says,

"On average, my scores were higher."

Is she correct?
Give a reason for your answer.
You **must** show your working.

[4 marks]

Question 12 continues on the next page

6

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12 (c) In the next quiz there are two multiple choice questions.

Each question has 3 options to choose from.

Stefan does not know the answers to the questions.

He chooses at random an answer to each question.

What is the probability that both his answers are correct?

[2 marks]

Answer _____

2

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

Do not write outside the box

13 **Candle company**

Maya runs a company that makes and sells candles.

13 (a) The mass of wax, in grams, needed to make a candle can be worked out using

$$W = v \times \left(\frac{100 - f}{100} \right) \times 0.83$$

W is the mass of wax, in grams

v is the volume of the candle, in cubic centimetres

f is the percentage of fragrance in the candle

The company makes a candle in the shape of a cylinder.

The candle has radius 4 cm and height 15 cm

The candle has 10% fragrance.

Work out the mass of wax, in kilograms, required to make 2500 of these candles.

[7 marks]

Answer _____ kg

Do not write outside the box

13 (b) The company sells large candles for £9.60 each, including 20% VAT.

How much VAT is there on each large candle?

[3 marks]

Answer _____

Question 13 continues on the next page

10

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13 (c) Katie works for Maya.

She has an annual income of £19 410

She has a personal allowance of £11 850

She pays 20% tax on the rest of her income.

How much income tax should she pay in a year?

[3 marks]

Answer £ _____

END OF QUESTIONS

3

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